VF Asia Footwear Laboratory Testing Manual

Test Method for Knot Slip Resistance	Test Method #: VF- B23 REF #: TM-R- 019 Page 1 of 1
Date Issued: May,2019 Revision #:1 Reason for revision: Rectify the unit of measure from kg to kgf for material strength.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the knot slip resistance of laces used in VF footwear.

REFERENCES: SATRA Test Method TM 195D

EQUIPMENT AND MATERIAL:

1. Tensile testing Machine with clamping jaws or with pneumatic grips
2. 5.0 kg weight
3. Vise grips or some other method of clamping to the lace
4. Laces

SAMPLE PREPARATION:

Condition the specimens at 23 ± 2 ℃ temperature and 55 ± 10% relative humidity for 24 hours prior to test.

REPETITIONS: Test six (6) samples

TEST PROCEDURE:
Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Tie a slipknot in the lace. There should be two ends of the lace in the knot that do not slip when pulled.
2. Attach the 5.0 kg weight to one end. Holding the other end, suspend the weight for a period of 10 seconds.
3. Once the lace has been set, place one fixed end and the sliding end into the clamps or pneumatic grips.
4. Set the speed on the tensile tester to 100 mm/min.
5. Pull the knot open and record the maximum force required to initiate knot slip.
6. Repeat the procedure in steps 1.0 through 4.0, for another five times for a total of six trials.

REPORT:

1. Average maximum force to initiate knot resistance: **kgf**
2. Also include i) lace description, ii) weight of the fixed weight used to set the knot, procedure.